

ADVANCED AND QUANTUM OPTICS — Paper 4

Model Solutions and Marking Scheme

Natural Sciences Tripos Part II — 2026

SECTION A

Question 1 [4 marks] — Fresnel Number and Airy Disk

The Fresnel number is $N_F = a^2/(\lambda d)$, where $a = D/2 = 0.25$ mm is the aperture radius.

At $d_1 = 5$ cm:

$$N_F = \frac{(2.5 \times 10^{-4})^2}{633 \times 10^{-9} \times 0.05} = \frac{6.25 \times 10^{-8}}{3.165 \times 10^{-8}} \approx 2.0.$$

$N_F \sim 1$, so we are in the **Fresnel (near-field)** regime. [1 mark]

At $d_2 = 5$ m:

$$N_F = \frac{(2.5 \times 10^{-4})^2}{633 \times 10^{-9} \times 5} = \frac{6.25 \times 10^{-8}}{3.165 \times 10^{-6}} = 0.020.$$

$N_F \ll 1$, so we are in the **Fraunhofer (far-field)** regime. [1 mark]

Airy disk radius: For a circular aperture of diameter D , the radius of the central Airy disk is

$$\rho = \frac{1.22 \lambda d}{D} = \frac{1.22 \times 633 \times 10^{-9} \times 5}{0.5 \times 10^{-3}} = \frac{3.86 \times 10^{-6}}{5 \times 10^{-4}} = 7.7 \times 10^{-3} \text{ m} \approx 7.7 \text{ mm}.$$

[2 marks]

Question 2 [4 marks] — Linewidth from Cross-Section

For a Lorentzian line, the lineshape function at line centre is $g(\nu_0) = 2/(\pi \Delta\nu)$. Substituting into the cross-section formula:

$$\sigma(\nu_0) = \frac{\lambda^2}{8\pi t_{\text{sp}}} \cdot \frac{2}{\pi \Delta\nu} = \frac{\lambda^2}{4\pi^2 t_{\text{sp}} \Delta\nu}.$$

Solving for $\Delta\nu$:

$$\Delta\nu = \frac{\lambda^2}{4\pi^2 t_{\text{sp}} \sigma(\nu_0)} = \frac{(780 \times 10^{-9})^2}{4\pi^2 \times 25 \times 10^{-9} \times 1.5 \times 10^{-17}}.$$

[2 marks]

Numerically:

$$\Delta\nu = \frac{6.084 \times 10^{-13}}{4\pi^2 \times 3.75 \times 10^{-25}} = \frac{6.084 \times 10^{-13}}{1.480 \times 10^{-23}} \approx 4.1 \times 10^{10} \text{ Hz} \approx 41 \text{ GHz}.$$

[1 mark]

Comment: The natural linewidth is

$$\Delta\nu_{\text{nat}} = \frac{1}{2\pi t_{\text{sp}}} = \frac{1}{2\pi \times 25 \times 10^{-9}} \approx 6.4 \text{ MHz}.$$

The deduced linewidth ($\sim 41 \text{ GHz}$) is several orders of magnitude larger than the natural linewidth, so the transition cannot be radiatively broadened alone. The actual line is dominated by other broadening mechanisms (Doppler broadening if a thermal gas, or collisional broadening, or inhomogeneous broadening in a solid). The given cross-section is therefore consistent with a much broader inhomogeneous line. [1 mark]

Question 3 [4 marks] — Sinusoidal Grating in a 2f System

The grating transmittance can be rewritten using $\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$:

$$t(x) = \frac{1}{2} + \frac{1}{4}e^{j2\pi x/\Lambda} + \frac{1}{4}e^{-j2\pi x/\Lambda}.$$

The Fourier transform contains three delta-function components: a DC term at $\nu_x = 0$ (amplitude $1/2$), and two side terms at $\nu_x = \pm 1/\Lambda$ (each of amplitude $1/4$). [1 mark]

By the Fourier-transform property of a lens, each spatial-frequency component appears as a bright spot in the back focal plane at position $x = \lambda f \nu_x$. So:

- $\nu_x = 0 \rightarrow x = 0$ (central spot).
- $\nu_x = \pm 1/\Lambda \rightarrow x = \pm \lambda f / \Lambda$.

Numerically:

$$\Delta x = \frac{\lambda f}{\Lambda} = \frac{500 \times 10^{-9} \times 0.20}{20 \times 10^{-6}} = \frac{1.0 \times 10^{-7}}{2.0 \times 10^{-5}} = 5.0 \times 10^{-3} \text{ m} = 5 \text{ mm}.$$

So spots appear at $x = 0$ and $x = \pm 5 \text{ mm}$. [2 marks]

Relative intensities: The intensity is the squared modulus of each Fourier amplitude. The DC spot has amplitude $1/2$ (intensity $1/4$); each side spot has amplitude $1/4$ (intensity $1/16$). Hence the central spot is $4\times$ brighter than each side spot, and the intensity ratio is

$$I_{\text{central}} : I_{\pm 1} = 4 : 1 : 1.$$

[1 mark]

Question 4 [4 marks] — Confocal Resonator

For a confocal resonator with $d = R$:

$$g_1 = g_2 = 1 - \frac{d}{R} = 0.$$

[1 mark]

The product $g_1 g_2 = 0$, which lies on the boundary of the stable region $0 \leq g_1 g_2 \leq 1$. The confocal configuration is marginally stable (in fact it is the most stable point in a practical sense — small misalignments do not destabilise it). **[1 mark]**

The waist radius of the fundamental Gaussian mode in a symmetric resonator is

$$W_0 = \sqrt{\frac{\lambda}{\pi} \left[\frac{d(2R - d)}{4} \right]^{1/2}} \quad (\text{general symmetric}),$$

which for the confocal case ($d = R$) simplifies to

$$W_0 = \sqrt{\frac{\lambda d}{2\pi}}.$$

[1 mark]

Numerically:

$$W_0 = \sqrt{\frac{1064 \times 10^{-9} \times 0.50}{2\pi}} = \sqrt{8.467 \times 10^{-8}} = 2.91 \times 10^{-4} \text{ m} \approx 0.29 \text{ mm}.$$

The corresponding Rayleigh range is $z_0 = \pi W_0^2 / \lambda = d/2 = 25 \text{ cm}$, as expected for a confocal resonator (the Rayleigh range equals half the mirror separation). **[1 mark]**

SECTION B

Question 5 [17 marks] — Spatial Filtering in a $4f$ Processor

(a) [4 marks]

Fourier-transform property of a thin lens: A field $f(x, y)$ placed in the front focal plane of a thin lens of focal length f produces, in the back focal plane, a field proportional to the spatial Fourier transform of $f(x, y)$, evaluated at spatial frequencies $(\nu_x, \nu_y) = (x/(\lambda f), y/(\lambda f))$:

$$g(x, y) = h_l F\left(\frac{x}{\lambda f}, \frac{y}{\lambda f}\right),$$

where $h_l = (j/(\lambda f)) \exp(j2kf)$. [1.5 marks]

$4f$ system as imaging: The first lens produces in its back focal plane the Fourier transform $F(x_1/(\lambda f), y_1/(\lambda f))$. This plane is also the front focal plane of the second lens. The second lens then takes the Fourier transform of *this* field. Since the Fourier transform of a Fourier transform is the original function with arguments inverted ($\mathcal{F}\{F(\nu)\} = f(-x)$), the field at the output plane is proportional to $f(-x, -y)$ — an inverted copy of the input. [2 marks]

This shows the system is a unit-magnification imaging system (with inversion). The key advantage over direct imaging is that the intermediate Fourier plane gives access to the spatial-frequency content of the image, where filters can be inserted. [0.5 marks]

(b) [4 marks]

In the Fourier plane, the position (x_1, y_1) corresponds to the spatial frequency

$$(\nu_x, \nu_y) = \left(\frac{x_1}{\lambda f}, \frac{y_1}{\lambda f}\right).$$

[1 mark]

The opaque stop blocks the region $\sqrt{x_1^2 + y_1^2} < r_0$, which corresponds to spatial frequencies satisfying

$$\sqrt{\nu_x^2 + \nu_y^2} < \frac{r_0}{\lambda f}.$$

[1 mark]

Edge enhancement: A circular aperture is uniform in its interior, which corresponds to mainly low spatial frequencies. The sharp edges, however, are localised in real space and therefore extend to high spatial frequencies in the Fourier domain. Removing the low frequencies leaves only the high-frequency content, which corresponds to the rapid intensity changes at the edges. The resulting image shows a bright outline around the aperture and a dark interior — an edge-enhanced version of the original. [2 marks]

(c) [5 marks]

The grating decomposes as

$$f(x, y) = \frac{1}{2} + \frac{1}{4}e^{j2\pi x/\Lambda} + \frac{1}{4}e^{-j2\pi x/\Lambda}.$$

Its 2D Fourier transform is

$$F(\nu_x, \nu_y) = \frac{1}{2}\delta(\nu_x)\delta(\nu_y) + \frac{1}{4}\delta(\nu_x - 1/\Lambda)\delta(\nu_y) + \frac{1}{4}\delta(\nu_x + 1/\Lambda)\delta(\nu_y).$$

[1.5 marks]

So three spots appear in the Fourier plane:

- Central spot at $(x_1, y_1) = (0, 0)$ with amplitude $1/2$.
- Side spots at $(x_1, y_1) = (\pm\lambda f/\Lambda, 0)$ with amplitude $1/4$ each.

[1 mark]

The opaque stop blocks only the central spot. The remaining field at the Fourier plane is

$$F'(\nu_x, \nu_y) = \frac{1}{4}\delta(\nu_x - 1/\Lambda)\delta(\nu_y) + \frac{1}{4}\delta(\nu_x + 1/\Lambda)\delta(\nu_y).$$

The second lens Fourier-transforms this back, giving a field at the output proportional to

$$g(x, y) \propto \frac{1}{4}e^{j2\pi x/\Lambda} + \frac{1}{4}e^{-j2\pi x/\Lambda} = \frac{1}{2}\cos(2\pi x/\Lambda).$$

[1.5 marks]

The resulting intensity is

$$I(x, y) \propto \cos^2(2\pi x/\Lambda) = \frac{1}{2}[1 + \cos(4\pi x/\Lambda)].$$

This is *not* uniform — it is a fringe pattern at half the original period. The intended interpretation: the average intensity is uniform (the DC level $1/2$ in arbitrary units), but the pattern oscillates spatially. The frequency-doubling occurs because the cosine grating, with its DC component removed, becomes a cos pattern whose intensity (square) doubles the spatial frequency. The mean intensity equals $I_0/8$ relative to the input (factor $1/4$ from each side-band amplitude squared, times 2 for both side bands). [1 mark]

(d) [4 marks]

If the filter passes only the spot at $(\nu_x = 1/\Lambda, \nu_y = 0)$, the field at the Fourier plane is

$$F'(\nu_x, \nu_y) = \frac{1}{4}\delta(\nu_x - 1/\Lambda)\delta(\nu_y).$$

[1.5 marks]

The second lens Fourier-transforms this, giving an output field proportional to

$$g(x, y) \propto \frac{1}{4}\exp(j2\pi x/\Lambda).$$

This is a tilted plane wave (a single Fourier component in pure form). The output intensity is uniform: $|g|^2 \propto 1/16$. [1.5 marks]

Physical interpretation: The filter has selectively extracted a single spatial-frequency component of the original image, converting a periodic intensity pattern into a uniform field carrying a tilted phase. This demonstrates the power of Fourier-plane filtering: by selecting which spots to pass or block in the Fourier plane, one can isolate, enhance, or suppress specific spatial-frequency content of an image. The same principle underlies dark-field microscopy, phase contrast microscopy, and matched-filter pattern recognition.

[1 mark]

Question 6 [17 marks] — Holography of a Point Source

(a) [4 marks]

Expanding the modulus squared:

$$|U_o + U_r|^2 = (U_o + U_r)(U_o^* + U_r^*) = |U_o|^2 + |U_r|^2 + U_o^*U_r + U_oU_r^*.$$

[1.5 marks]

So the transmittance is

$$t(x, y) = t_0 + \beta(|U_o|^2 + |U_r|^2 + U_o^*U_r + U_oU_r^*),$$

which contains four distinct contributions:

- (i) $|U_r|^2 = |A_r|^2$ — a uniform background (the reference intensity).
- (ii) $|U_o|^2$ — the (usually weak) object self-intensity.
- (iii) $\beta U_oU_r^*$ — the cross-term carrying the object's complex amplitude (multiplied by U_r^*).
- (iv) $\beta U_o^*U_r$ — the conjugate cross-term carrying U_o^* .

[1.5 marks]

The term $U_oU_r^*$ contains the full information about U_o (both amplitude and phase), and is the term used to reconstruct the object wave on illumination by U_r . The term $U_o^*U_r$ is responsible for the so-called **conjugate (twin) image** — it produces a wave that converges to the mirror-image position of the original object on the opposite side of the hologram plane. [1 mark]

(b) [5 marks]

Illuminating the developed hologram with the reference wave U_r produces a transmitted field

$$U_{\text{out}}(x, y) = U_r(x, y) t(x, y) = t_0U_r + \beta U_r|U_o|^2 + \beta U_r|U_r|^2 + \beta U_r^2U_o^* + \beta|U_r|^2U_o.$$

[2 marks]

The five contributions (collected by physical interpretation) are:

- (i) $(t_0 + \beta|U_r|^2)U_r$ — the directly transmitted reference wave (proceeds in the direction of U_r , i.e. at angle θ to the z -axis).
- (ii) $\beta U_r|U_o|^2$ — a low-amplitude correction propagating roughly in the direction of U_r .
- (iii) $\beta|U_r|^2U_o = \beta A_r^2 U_o$ — **the reconstructed object wave**, identical to the original U_o up to a constant. This wave appears to diverge from the location of the original object, producing a virtual image.
- (iv) $\beta U_r^2U_o^* = \beta A_r^2 e^{-j2k \sin \theta x} U_o^*$ — the conjugate (twin) image, which carries the phase-conjugate of the object wave together with an additional linear phase $\exp(-j2k \sin \theta x)$, meaning it propagates at angle -2θ relative to the reconstructed wave.

[2 marks]

Off-axis separation: Because the reference wave is incident at angle θ , the directly transmitted reference beam exits at angle θ , the reconstructed object wave continues in

its original direction (along the z -axis if the object was on-axis), and the conjugate image is deflected to angle $-\theta$. By choosing θ large enough, all three beams are angularly separated and the observer can view the reconstructed virtual image without interference from the other contributions. This is the essence of off-axis (Leith–Upatnieks) holography. **[1 mark]**

(c) [5 marks]

Substituting the point-source field into the cross-term $\beta U_r^* U_o$ (the term producing the conjugate image, which is the lens-like one):

$$U_r^* U_o = A_r^* \exp(jk \sin \theta x) \cdot \frac{A_o}{z_0} \exp(jk z_0) \exp\left[\frac{jk(x^2 + y^2)}{2z_0}\right].$$

Combining the constants into $C \equiv (A_r^* A_o / z_0) \exp(jk z_0)$:

$$U_r^* U_o = C \exp\left[\frac{jk(x^2 + y^2)}{2z_0}\right] \exp(jk \sin \theta x).$$

[2 marks]

The first exponential is a parabolic phase factor that is identical to the transmission function of a thin lens of focal length f_H , where the standard thin-lens transmittance is $\exp[-jk(x^2 + y^2)/(2f_H)]$. Comparing signs:

$$\frac{1}{2z_0} = -\frac{1}{2f_H}, \quad \implies \quad f_H = -z_0.$$

The negative sign indicates a **diverging** lens of focal length z_0 . Equivalently, the conjugate term $U_r U_o^*$ would give $f_H = +z_0$ — a converging lens. **[1.5 marks]**

The linear phase factor $\exp(jk \sin \theta x)$ is a tilt: it shifts the centre of the lens-like pattern laterally by an amount proportional to $\sin \theta$. More precisely, when we complete the square, the centre of the parabolic phase is displaced from the optical axis by

$$x_{\text{centre}} = -z_0 \sin \theta.$$

The hologram thus records a laterally offset Fresnel zone plate, which acts as an off-axis lens. **[1.5 marks]**

(d) [3 marks]

When illuminated by the reference wave, this off-axis Fresnel zone plate behaves as a thin lens of focal length $f_H = z_0$. Light from the reference (effectively a plane wave) is focused by the converging part of the zone plate to a real image at distance $f_H = z_0$ behind the hologram, on the opposite side from the original object. Simultaneously, the diverging part of the zone plate produces a virtual image at distance z_0 in front of the hologram — at exactly the position of the original point source. **[1.5 marks]**

In-line vs. off-axis: In Gabor's original in-line geometry ($\theta = 0$), the reference and object waves both propagate along the z -axis, so the directly transmitted reference, the reconstructed object wave (virtual image at $-z_0$), and the conjugate (real image at $+z_0$) all overlap spatially. The viewer sees the desired virtual image superimposed on the unwanted

out-of-focus real image and the bright zero-order beam. By tilting the reference wave to angle θ , Leith and Upatnieks gave each of these three contributions a different spatial direction, so they can be physically separated and observed independently. [**1.5 marks**]